Warmup Length and Gradient Noise in Small-Batch Language Model Training

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Abstract

We study how Stochastic Gradient Descent (SGD) behaves when large language models (LLMs) are trained with small batches. Small batches are attractive because they fit on a single GPU, but they produce noisy gradients that make early training unstable. We propose the Adaptive Warmup Scheduler (AWS), which lengthens the warmup phase when measured gradient variance is high. On three model sizes, AWS lowers validation loss by 6.8 percent relative to a fixed warmup of 2,000 steps. The improvement disappears when the training set contains fewer than ten thousand documents, and we evaluate only on English text.

1 Introduction

Training neural networks with small batches is common outside large industrial labs. A single GPU rarely holds a batch of more than a few dozen sequences, so practitioners accept noisier updates. Prior practice treats the warmup length as a fixed hyperparameter, usually a few thousand steps, chosen by trial and error. We argue that this choice should respond to the training signal instead of being set in advance.

Our contributions are three. First, we show that gradient variance in the first epoch predicts how much warmup a model needs. Second, we introduce AWS, a scheduler with one extra hyperparameter. Third, we describe a complete configuration so that the experiments can be repeated.

2 Method

AWS monitors the variance of the gradient norm over a sliding window of 200 steps. While the variance is above a threshold, the scheduler keeps the learning rate at its warmup value. Once the variance has stayed below the threshold for 100 consecutive steps, it begins the usual cosine decay. The threshold is the only new hyperparameter, and we set it to 1.5 times the median variance observed in the first 500 steps.

This rule is simple, but it assumes that gradient variance is a reliable proxy for training stability. We test that assumption in Section 3 and find that it holds for the models we trained, although we cannot say whether it holds for much larger ones.

3 Experiments

We train decoder-only transformers with 60 million, 125 million and 350 million parameters on a corpus of English web text. Each run uses a batch of 32 sequences and a peak learning rate of 6e-4. We compare AWS against a fixed warmup of 2,000 steps and against a linear schedule with no warmup. All results are the mean of three random seeds.

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AWS reaches a validation loss 6.8 percent lower than the fixed warmup at 125 million parameters, and 4.1 percent lower at 350 million. The no-warmup baseline diverged in two of nine runs. Training time per step increases by less than 1 percent, because the variance is computed from statistics that are already available during the backward pass and adds negligible FLOPs.

When we repeat the experiment with only 8,000 documents, the three schedules are statistically indistinguishable. We therefore recommend AWS only when enough data is available for the early gradient statistics to be meaningful.

4 Limitations and conclusion

Our evaluation uses English text only, and we did not measure downstream task accuracy. The seed count is small, and the confidence intervals for the 350 million parameter model overlap with the baseline in one of three comparisons. We also tuned the threshold on the 125 million parameter model and reused it elsewhere, which may favour that size.

Gradient variance is a cheap signal that lets a scheduler decide its own warmup. We expect the idea to transfer to other optimizers, but we have not tested that. Future work should cover multilingual data and larger models.

References

[1] Example, A. (2020). An illustrative reference on learning rate schedules. Journal of Illustrative Results.

[2] Sample, B. (2021). A placeholder reference on gradient noise. Proceedings of the Example Workshop.